

Transform Calculus

(MA 20202)

Assignment-1

1. (a) Show that the Laplace transform of the function $\frac{1}{\sqrt{t}}$ is $\sqrt{\frac{\pi}{s}}$.
(b) State the sufficient condition for the function $f(t), t \geq 0$ to have the Laplace transform. Give one example to show that these conditions are not the necessary conditions for $f(t)$ to have Laplace transform.
2. (a) If the Laplace transform of the function $f(t)$ is $F(s)$ and c is a positive constant, show that the Laplace transform of the function $f(ct)$ is $\frac{F(s/c)}{c}$.
(b) Find the Laplace transform of $\cos(\omega t)$ (You can use that the Laplace transform of $\cos t$ is $\frac{s}{s^2+1}$).
3. Find the Laplace transform of the following functions-
 - (a) $\cos^2\left(\frac{1}{2}\pi t\right)$,
 - (b) $t^3 e^{-3t}$,
 - (c) $e^{-t/2} u(t-2)$,
 - (d) $(t-a)^n u(t-a)$.
4. Show that the Laplace transform of a piecewise continuous function $f(t)$ with period p is

$$\frac{1}{1 - e^{-ps}} \int_0^p e^{-st} f(t) dt \quad (s > 0).$$

5. Prove that the Laplace transform of

- (a) $t \cos \omega t$ is $\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$,
- (b) $t \sinh \omega t$ is $\frac{2\omega s}{(s^2 - \omega^2)^2}$.

(Hint: Use the results of Laplace transform of derivatives).

6. If f is continuous, except for an ordinary discontinuity (finite jump) at some $t = a (> 0)$, satisfies the growth restriction, $f'(t)$ is piecewise continuous on every finite interval in $[0, \infty)$ except at $t = a$. Then show that the Laplace transform of $f'(t)$ is

$$sF(s) - f(0) - [f(a+0) - f(a-0)]e^{-as}$$

(where $F(s)$ is Laplace transform of $f(t)$.)

7. If $F(s) = \frac{A_1}{(s-a)^m} + \frac{A_2}{(s-a)^{m-1}} + \cdots + \frac{A_m}{s-a} + \frac{B_1}{s-b_1} + \frac{B_2}{s-b_2} + \cdots + \frac{B_n}{s-b_n}$, then find the inverse Laplace transform of $F(s)$.
8. Express the following functions in terms of Heaviside unit step functions:-

$$(a) f(t) = \begin{cases} t^2 & \text{if } 0 < t < 2 \\ 4t & \text{if } t > 2 \end{cases}$$

$$(b) f(t) = \begin{cases} \sin t & \text{if } 0 < t < \pi \\ \sin 2t & \text{if } \pi < t < 2\pi \\ \sin 3t & \text{if } t > 2\pi \end{cases}$$

9. Find the Laplace transform of the following functions-

$$(a) f(t) = \begin{cases} 5 & \text{if } 0 < t < 7 \\ 0 & \text{if } t \geq 7 \end{cases}$$

$$(b) f(t) = \begin{cases} \sin t & \text{if } \pi/2 < t < \pi \\ 0 & \text{if } t \leq \pi/2, t \geq \pi \end{cases}$$

10. Find the Laplace transform of the following functions-

$$(a) t^2 \sin 3t,$$

$$(b) \frac{\sin at}{t},$$

$$(c) 4t * e^{-2t}.$$

11. (a) If the Laplace transform of $f(t)$ is $F(s)$, prove that the Laplace transform of $t^n f(t)$ is $(-1)^n F^{(n)}(s)$ (where $F^{(n)}(s)$ is the n -th derivative of $F(s)$).
- (b) Find the Laplace transform $t^n e^{kt}$.

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